Machine Learning — Lecture 3

Linear Regression, Optimization, and Regularization

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Lecture 3 at a Glance

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Linear Regression, Optimization, and Regularization

Course Learning Outcomes

CLO3: Construct and analyze machine learning models by applying algorithmic principles and internal computations.

CLO4: Apply supervised and unsupervised learning techniques to solve practical problems and interpret their outcomes.

Lecture purpose

- Introduce the first full supervised-learning model.
- Connect prediction, residuals, loss, and optimization.
- Derive how gradient descent updates model parameters.
- Explain how regularization controls model flexibility.
- Evaluate regression models using appropriate metrics.

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💡 Main idea: linear regression is simple, but it contains the complete learning pattern: **model → prediction → error → loss → optimization → evaluation**.

Learning Objectives

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By the end of Lecture 3, you will be able to:

1. Formulate regression as a supervised-learning problem.
2. Represent simple and multiple **linear regression** mathematically.
3. Compute predictions, residuals, and squared errors.
4. Explain Mean Squared Error as an objective function.
5. Describe how gradient descent reduces loss.
6. Perform one manual parameter update for a simple linear model.
7. Distinguish batch, stochastic, and mini-batch gradient descent.
8. Explain overfitting through model flexibility and polynomial features.
9. Explain Ridge and Lasso regularization conceptually.
10. Evaluate regression models using MAE, MSE, RMSE, and R^2, and justify a model choice on validation data.

Lecture Roadmap

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From a line to a learning algorithm

Learning phase	Parts	Core question	Student output
1. Formulate the model	**I–II**	What are we predicting, and how does a linear model produce a prediction?	Define X, y, $\hat{y}$, coefficients, and residuals.
2. Learn the parameters	**III–IV**	How do we measure errors and reduce a loss?	Calculate MSE and one gradient update.
3. Control complexity	**V–VI**	When do flexible models overfit, and how does regularization help?	Reason about polynomial features, Ridge, and Lasso.
4. Evaluate and apply	**VII–VIII**	How do we judge a regression model on unseen cases?	Compare validation metrics and justify a model choice.

💡 **Practice & consolidation:** calculate predictions, recognize overfitting and common mistakes, and explain defensible modeling decisions.

Part I

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Regression as Supervised Learning

Regression predicts a **continuous numerical target** from one or more **input features**.

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What is a Regression Problem?

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Predicting a numerical quantity

In supervised learning, we observe examples: $(x_i, y_i), \quad i = 1, \ldots, n$

For regression:

- x_i is the input feature vector;
- y_i is a real-valued target;
- the model predicts a number: $\hat{y}_i = f(x_i)$

Examples

Problem	Input features	Target
Predict apartment rent	area, district, rooms, age	monthly rent
Predict delivery time	distance, traffic, weather	minutes
Predict exam score	study hours, attendance, previous GPA	score
Predict customer spending	orders, visits, recency	spending amount

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💡 Regression is not about assigning a class. It is about estimating a **numerical value**.

The Learning Task

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Approximate an unknown relationship

The true relationship between features and target is usually unknown:

$$y = f(x) + \varepsilon$$

where:

- $f(x)$ is the systematic pattern;
- ε is noise, measurement error, or unobserved influence.

A model attempts to learn an approximation:

$$\hat{y} = \hat{f}(x)$$

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Goal

Find a model that predicts well on **new examples**, not only on the training data.

> A rent model should work for future apartments, not only for the apartments used during training.

Dataset Representation

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Feature matrix and target vector

A regression dataset can be represented as:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}, \qquad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}.$$

Each row of X is one observation.

Each column of X is one feature.

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Prediction target

The model learns a function from features to target:

$$X \to y$$

Example Dataset

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Predicting final exam score

Student	Study Hours	Attendance (%)	Previous GPA	Final Score
A	3	60	2.1	55
B	6	75	2.8	68
C	9	88	3.2	81
D	12	95	3.6	90

Possible supervised-learning formulation

- Features: study hours, attendance, previous GPA
- Target: final score
- Task: regression
- Model output: predicted final score

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💡 The target column should not be included as an input feature.

Residuals

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How wrong is the prediction?

For each observation, the residual is:

$$e_i = y_i - \hat{y}_i$$

Interpretation:

Residual	Meaning
$e_i > 0$	model underpredicted
$e_i < 0$	model overpredicted
$e_i = 0$	exact prediction

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Why residuals matter

Residuals are the raw material for loss functions and diagnostic plots.

> If residuals show a **systematic pattern**, the model may be too simple or missing an important feature.

Part II

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Simple Linear Regression

Simple **linear regression** uses one feature to predict one continuous target using a straight line.

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Simple Linear Model

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One feature, one target

Simple **linear regression** assumes: $\hat{y} = w_0 + w_1 x$

Where:

- w_0 is the intercept;
- w_1 is the slope;
- x is the input feature;
- $\hat{y}$ is the predicted target.

Interpretation

Parameter	Meaning
w_0	predicted value when $x = 0$
w_1	expected change in $\hat{y}$ for one-unit increase in x

💡 The model is linear in the parameters, not necessarily always sufficient for the data.

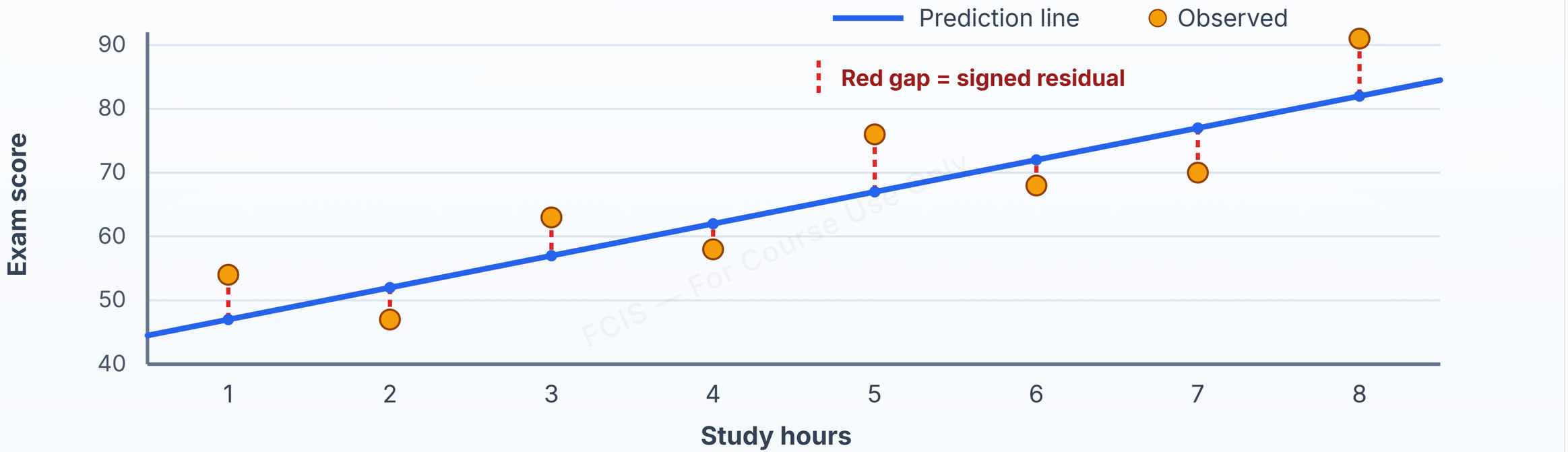
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Visual Interpretation

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A fitted line and its residuals

Each dot represents a student. The blue line gives the prediction $\hat{y}_i$; a vertical red segment shows the **signed residual** $e_i = y_i - \hat{y}_i$.



A residual above the line means the model **underpredicted**. Systematic residual patterns suggest the model may be missing structure.

Worked Prediction Example

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Compute $\hat{y}$ and residuals

Suppose the model is:

$$\hat{y} = 40 + 4x$$

where x is study hours.

For a student who studied $x = 6$ hours:

$$\hat{y} = 40 + 4(6) = 64$$

If the actual score is:

$$y = 68$$

then the residual is:

$$e = y - \hat{y} = 68 - 64 = 4$$

> The model underpredicted by 4 points.

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Multiple Linear Regression

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Many features, one target

With p features, the model becomes:

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_p x_p$$

Vector form:

$$\hat{y} = w^\top x + b$$

Where:

- w is a vector of feature weights;
- b is the intercept;
- x is the feature vector.

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Example

$$\widehat{\text{score}} = 30 + 3(\text{hours}) + 0.2(\text{attendance}) + 5(\text{GPA})$$

Interpreting Coefficients

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What does a weight mean?

In a linear model: $\hat{y} = w_0 + w_1 x_1 + \cdots + w_p x_p$

the coefficient w_j is interpreted as the expected change in prediction when x_j increases by one unit, holding other features fixed.

Cautions

Issue	Why interpretation becomes difficult
Different feature scales	coefficients cannot be compared directly
Correlated features	effects are difficult to separate
Nonlinear relationships	linear coefficient may oversimplify
Interactions	effect of one feature may depend on another
Categorical encoding	interpretation depends on reference category

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💡 **Coefficient** interpretation requires both mathematics and domain understanding.

Model Assumptions

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What linear regression assumes approximately

Classical **linear regression** is most appropriate when:

- the relationship is approximately linear;
- residuals have mean near zero;
- residual variance is reasonably stable;
- observations are independent;
- features are measured appropriately;
- extreme outliers are not dominating the fit.

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In machine learning practice

We often focus less on formal statistical inference and more on predictive performance.

But assumptions still matter because they explain **when the model may fail**.

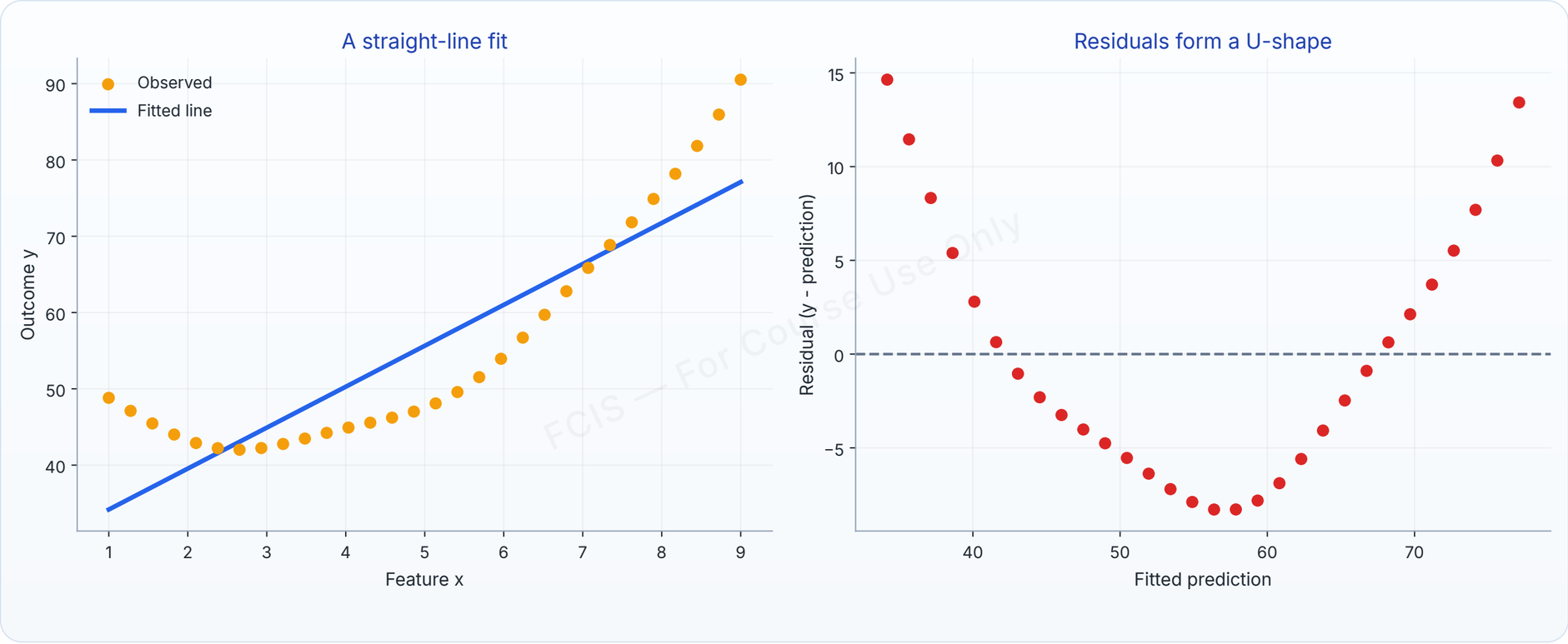
> A highly curved relationship may require feature transformations, polynomial terms, or a different model.

Residual Diagnostics

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A residual pattern can reveal a missing nonlinear relationship

This **illustrative dataset** shows what happens when a straight line is fitted to a curved relationship.



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💡 **Takeaway.** The curved **residual** pattern suggests the linear model missed structure. Residuals scattered randomly around zero would be more reassuring.

Part III

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Loss Functions

A **loss function** turns prediction errors into a numerical objective that the learning algorithm can minimize.

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Why We Need a Loss Function

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From an individual error to a learning objective

Many possible lines can be drawn through the same data. To learn which is better, we need a numerical criterion.

Concept	Definition	Example
Residual	Signed difference for one case: $e_i = y_i - \hat{y}_i$	$e_i = -4$ (overprediction)
Per-example loss	Nonnegative penalty for one case: $\ell_i = (y_i - \hat{y}_i)^2$	$\ell_i = 16$
Training objective	Aggregate of losses over the training examples	$MSE = \frac{1}{n}\sum_i \ell_i$

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Absolute error $|e_i|$ is another possible per-example loss; the choice of loss changes what a model considers a good fit.

💡 A **residual** keeps its sign; a **loss** scores one prediction; an **objective** aggregates losses to guide parameter learning.

Mean Squared Error

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The standard regression objective

Mean Squared Error is:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Why square the errors?

- negative and positive errors do not cancel;
- larger errors are penalized more strongly;
- the objective is smooth and convenient for **optimization**.

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But

MSE can be sensitive to outliers because large errors are squared.

Worked Example: MSE

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Three predictions

Observation	Actual y	Prediction $\hat{y}$	Error $y - \hat{y}$	Squared error
1	70	68	2	4
2	80	76	4	16
3	90	95	-5	25

Then:

$$MSE = \frac{4 + 16 + 25}{3} = 15$$

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Root Mean Squared Error:

$$RMSE = \sqrt{15} \approx 3.87$$

RMSE is in the same unit as the target, which often makes it easier to interpret.

Loss Surface

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Parameters determine loss

For a simple model:

$$\hat{y} = w_0 + w_1 x$$

the loss depends on the parameters:

$$L(w_0, w_1) = \frac{1}{n} \sum_{i=1}^{n} (y_i - (w_0 + w_1 x_i))^2$$

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Learning means finding parameters that reduce this loss.

> 💡 **Optimization** is the process of moving through parameter space toward lower loss.

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Gradient Descent

Gradient descent is a general method for learning parameters by repeatedly moving in the direction that reduces the loss.

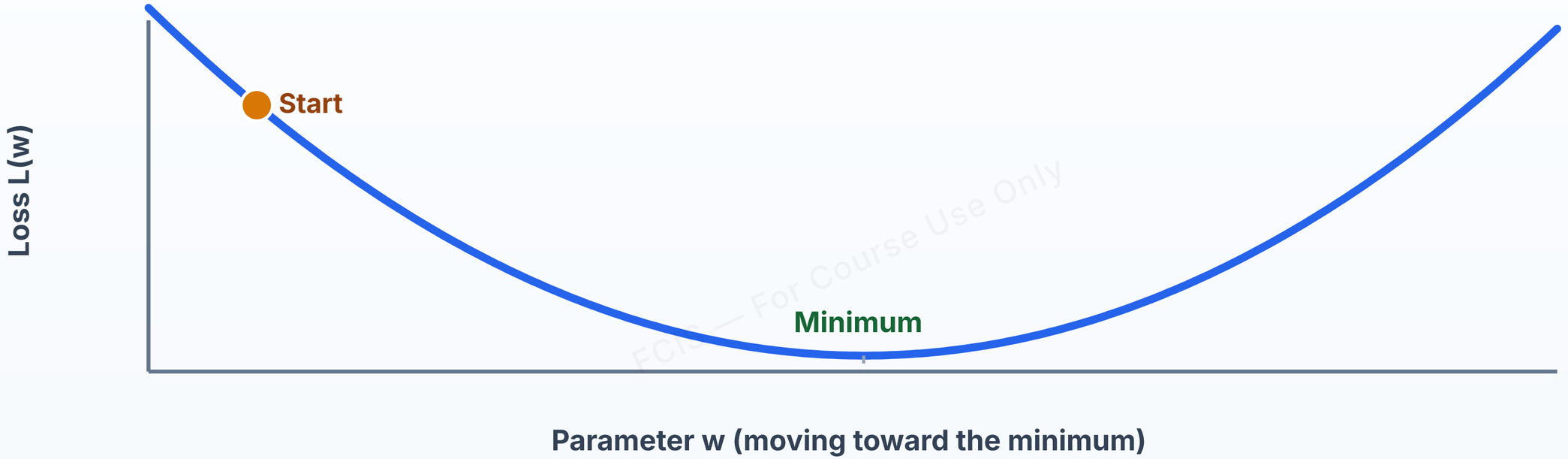
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Gradient Descent Intuition

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Follow the slope downhill

The derivative $\frac{dL}{dw}$ tells us how the loss changes with w. The update $w \leftarrow w - \alpha \frac{dL}{dw}$ moves **opposite** the slope. Advance the slide to reveal successive updates.

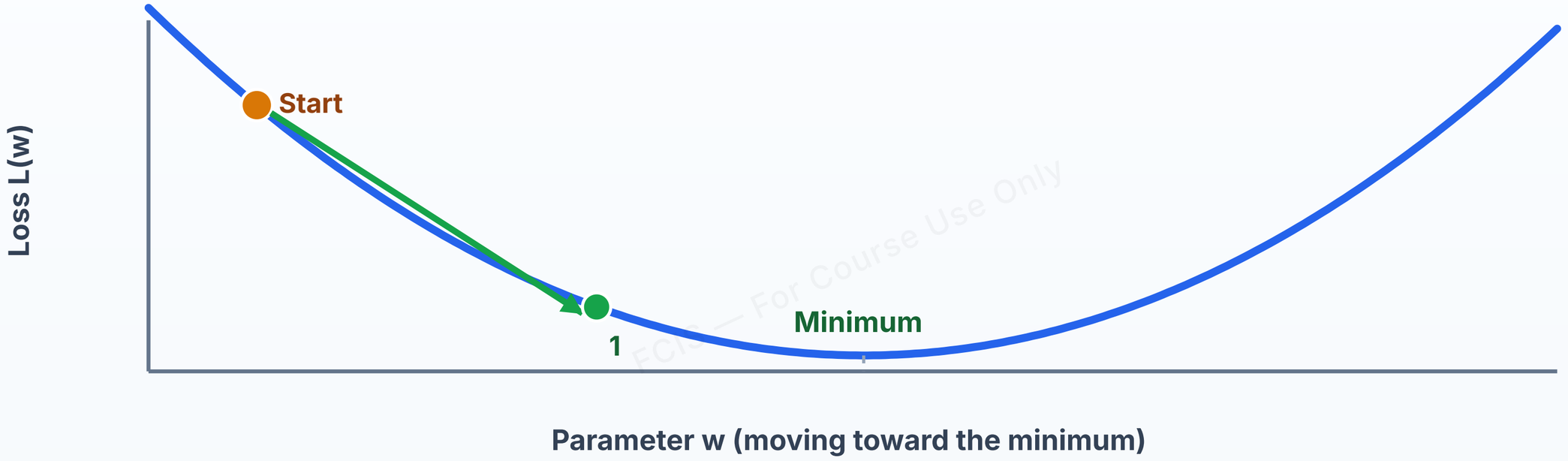


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Positive slope $\Rightarrow$ decrease w; negative slope $\Rightarrow$ increase w. The goal is **lower loss**, not simply a larger parameter.

Follow the slope downhill

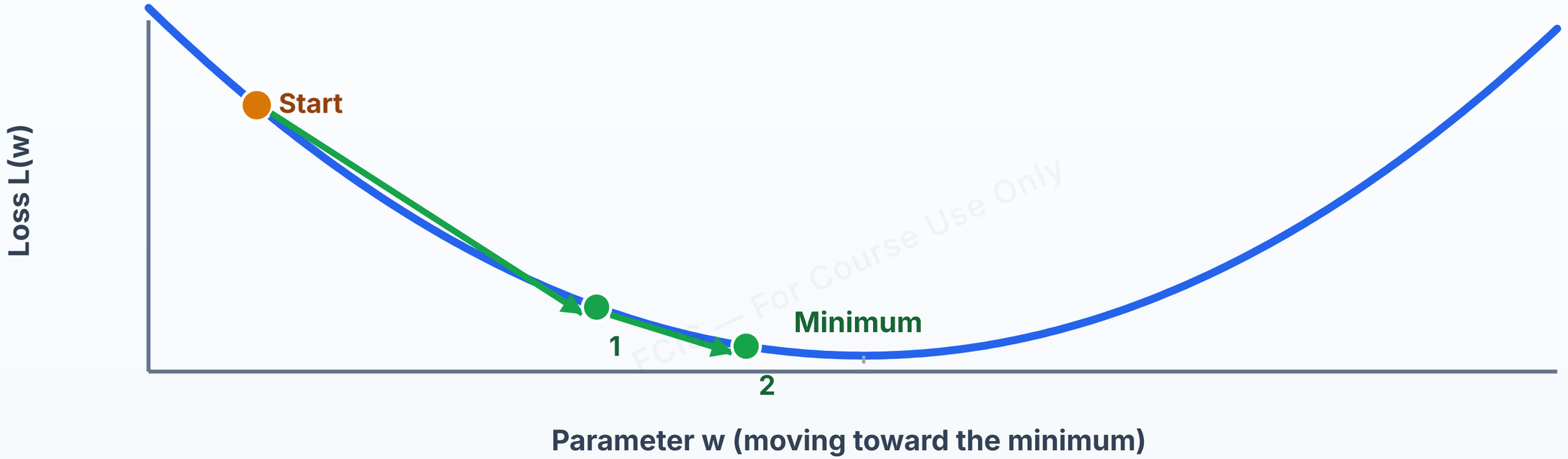
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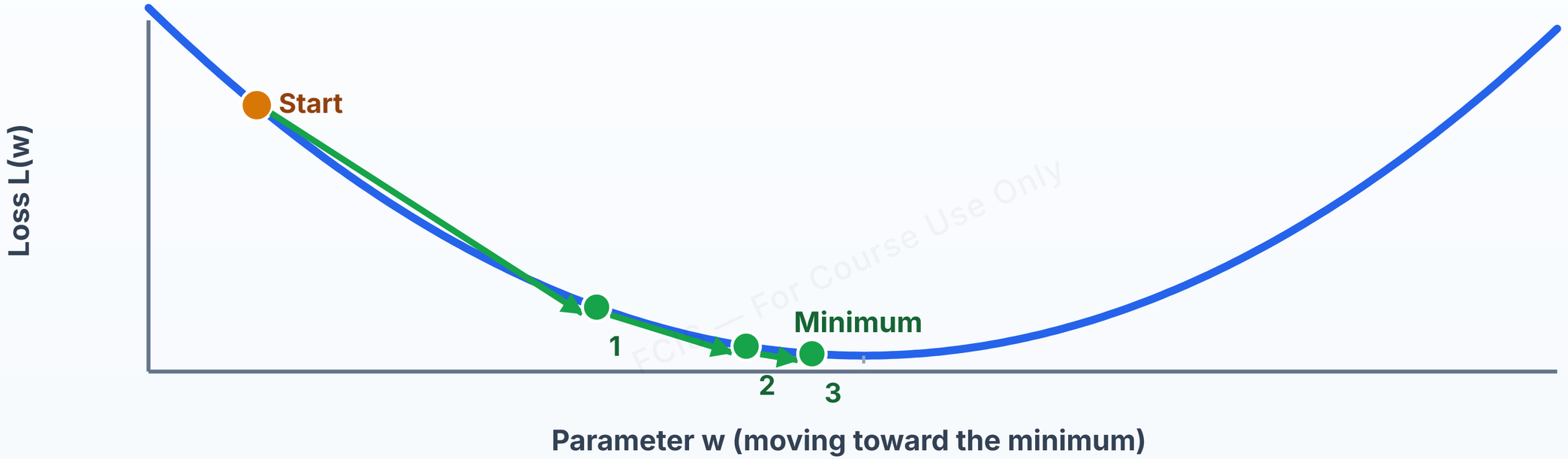
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Learning Rate

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Small, useful, or excessively large steps

The learning rate α sets the step size in $w \leftarrow w - \alpha \nabla_w L$.

Learning rate	Possible behavior
Too small	Converges very slowly
Appropriate	Moves toward a minimum
Too large	Overshoots; may oscillate or diverge

Smaller step — approaches minimum



Shorter moves settle near the bottom

Oversized step — overshoots



A large step can jump past the minimum

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💡 The picture is schematic: whether a rate is too large depends on the **loss curvature and feature scale**.

Gradient for Simple Linear Regression

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One data point

For one example: $\hat{y} = wx + b$

Use squared error: $L = (\hat{y} - y)^2$

Let the prediction error be: $r = \hat{y} - y$

Then: $\frac{\partial L}{\partial w} = 2rx$

and: $\frac{\partial L}{\partial b} = 2r$

Update:

$$w \leftarrow w - \alpha(2rx)$$

$$b \leftarrow b - \alpha(2r)$$

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Worked Gradient Update

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One example, one step — and check the loss

For $x = 3$, $y = 10$, and current parameters $w = 2$, $b = 1$:

$$\hat{y} = wx + b = 7, \qquad r = \hat{y} - y = -3, \qquad L = r^2 = 9.$$

The gradients of this **one-example squared loss** are:

$$\frac{\partial L}{\partial w} = 2rx = -18, \qquad \frac{\partial L}{\partial b} = 2r = -6.$$

Choose $\alpha = 0.02$ and update **both** parameters:

$$w_{\text{new}} = 2 - 0.02(-18) = 2.36, \qquad b_{\text{new}} = 1 - 0.02(-6) = 1.12.$$

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Check: $\hat{y}_{\text{new}} = 2.36(3) + 1.12 = 8.2$, so $L_{\text{new}} = (8.2 - 10)^2 = 3.24$.

Update	Prediction	Squared loss
Before	7.0	**9.00**
After ($\alpha = 0.02$)	8.2	**3.24**

A larger step, $\alpha = 0.1$, would predict **13**, leaving the squared loss at **9**. An update does not automatically improve the objective.

Batch Gradient Descent

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Use all training examples before updating

Batch **Gradient Descent** computes gradients using the entire training set:

$$\nabla L = \frac{1}{n} \sum_{i=1}^{n} \nabla L_i$$

Advantages

- stable direction;
- directly follows the full **loss surface**;
- good for small datasets.

Limitations

- expensive for very large datasets;
- one update requires scanning all examples;
- can be slow when n is large.

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> 💡 Batch gradient descent is conceptually clean but not always computationally efficient.

Stochastic Gradient Descent

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Update using one example at a time

Stochastic **Gradient Descent** updates parameters using one training example at a time.

Advantages

- fast updates;
- can handle large datasets;
- noisy updates may help escape shallow regions.

Limitations

- update direction is noisy;
- loss curve may fluctuate;
- **learning rate** scheduling becomes important.

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$$w \leftarrow w - \alpha \nabla L_i(w)$$

SGD trades stability for speed and scalability.

Mini-Batch Gradient Descent

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A practical compromise

Mini-batch **Gradient Descent** uses a small subset of examples at each update.

If a mini-batch B has m examples:

$$\nabla L_B = \frac{1}{m} \sum_{i \in B} \nabla L_i.$$

Why common in practice?

- more stable than SGD;
- less expensive than full batch;
- efficient on modern hardware;
- works naturally with neural networks.

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💡 Mini-batch gradient descent is often the default training strategy for large-scale models.

Comparing Gradient Descent Variants

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Stability vs. efficiency

Method	Data per update	Strength	Weakness
Batch GD	all examples	stable gradients	expensive for large data
SGD	one example	fast frequent updates	noisy trajectory
Mini-batch GD	small batch	practical compromise	still needs tuning

Conceptual comparison

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Batch: stable but expensive

SGD: fast but noisy

Mini-batch: balanced

When Optimization Fails

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Common causes

Gradient descent may behave poorly when:

- the learning rate is too large;
- the learning rate is too small;
- features are on very different scales;
- the loss surface is ill-conditioned;
- the model is too flexible for the data;
- gradients vanish or explode in deep networks.

Practical response

- scale features;
- monitor training and validation loss;
- tune learning rate;
- use mini-batches;
- add regularization;
- stop training when validation performance worsens.

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💡 **Optimization** behavior is strongly connected to preprocessing and model complexity.

Part V

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Model Flexibility and Overfitting

A model that fits training data extremely well may still generalize poorly.

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Underfitting and Overfitting

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Revisit generalization through regression curves

Lecture 1 introduced underfitting and overfitting conceptually. Here we use **regression curve flexibility** to see the mechanism.

Situation	Training error	Validation error	Regression interpretation
Underfitting	High	High	Straight line misses curvature
Useful fit	Moderate/low	Low	Model captures the stable trend
Overfitting	Very low	High	Highly flexible curve follows noise

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💡 Training error alone cannot select the polynomial degree. **Validation performance** tells us which curve transfers better to unseen data.

Polynomial Features

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Increasing model flexibility

A linear model in x may be too simple for curved data.

We can add **polynomial features**:

$$x, \quad x^2, \quad x^3, \ldots$$

Then:

$$\hat{y} = w_0 + w_1 x + w_2 x^2 + w_3 x^3$$

The model is still linear in the parameters w_j, but it can represent nonlinear curves in the original input.

Risk

More polynomial terms can reduce **training error** but increase overfitting.

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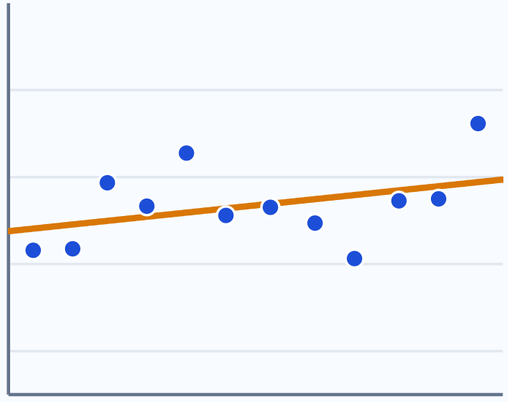
Flexibility Example

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Same observations, three regression curves

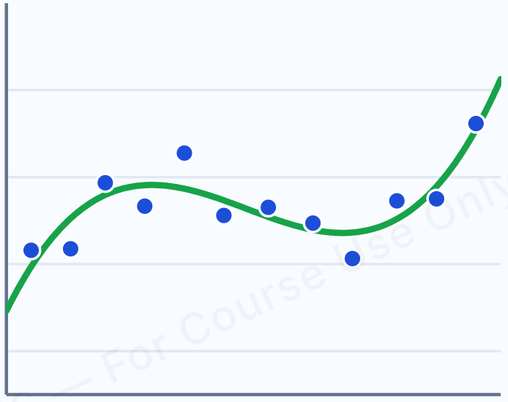
Each panel uses the **same illustrative noisy training observations** (blue dots). Only the polynomial degree changes.

Degree 1 — underfit



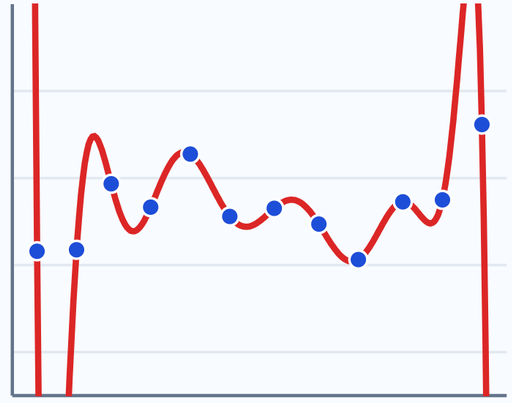
Misses the bend

Degree 3 — balanced



Captures the broad trend

Degree 11 — overfit



Chases individual noise

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Degree 11 can nearly interpolate the training data but still generalize poorly. Use **validation RMSE**, not visual complexity or training fit, to choose the degree.

Bias–Variance Intuition

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A qualitative view

The expected prediction error can be understood through:

- bias;
- variance;
- irreducible noise.

High bias

The model is too simple and misses the pattern.

High variance

The model is too sensitive to the training data.

> 💡 Increasing flexibility often decreases bias but increases variance.

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Part VI

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Regularization

Regularization discourages unnecessarily complex models by adding a penalty to the loss.

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Why Regularize?

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Control model complexity

The ordinary regression objective minimizes **training error**:

$$\text{Training Loss}$$

Regularization modifies the objective:

$$\text{Training Loss} + \lambda \times \text{Complexity Penalty}$$

Where:

- λ controls the strength of regularization;
- larger λ means stronger penalty;
- λ is a hyperparameter.

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💡 Regularization accepts a little more training error if it improves **generalization**.

Ridge Regression

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Penalize large squared coefficients

Ridge regression adds an L_2 penalty:

$$L_{\text{ridge}} = MSE + \lambda \sum_{j=1}^{p} w_j^2$$

Effect

- shrinks coefficients toward zero;
- usually keeps all features;
- reduces sensitivity to noise and multicollinearity;
- often improves **generalization**.

> Ridge is useful when many features may contribute a little.

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Lasso Regression

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Penalize absolute coefficient size

Lasso adds an L_1 penalty:

$$L_{\text{lasso}} = MSE + \lambda \sum_{j=1}^{p} |w_j|$$

Effect

- shrinks coefficients;
- can set some coefficients exactly to zero;
- performs a form of feature selection;
- useful when only a subset of features is expected to matter.

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> 💡 **Lasso** can simplify the model by removing some features from the prediction.

Ridge vs. Lasso

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Similar goal, different behavior

Aspect	Ridge	Lasso
Penalty	$\sum w_j^2$	$\sum \lvert w_j \rvert$
Coefficient effect	shrinks weights	can set weights to zero
Feature selection	usually no	yes, often
Useful when	many features contribute	few features matter
Stability	often stable with correlated features	may choose one of correlated features

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Both methods require selecting λ, usually using validation or cross-validation.

Regularization Strength

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What happens as λ changes?

λ	Expected behavior
0	ordinary linear regression
small	mild shrinkage
moderate	improved generalization possible
very large	coefficients strongly suppressed; underfitting risk

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Selection principle

Choose λ using validation performance, not **training error** alone.

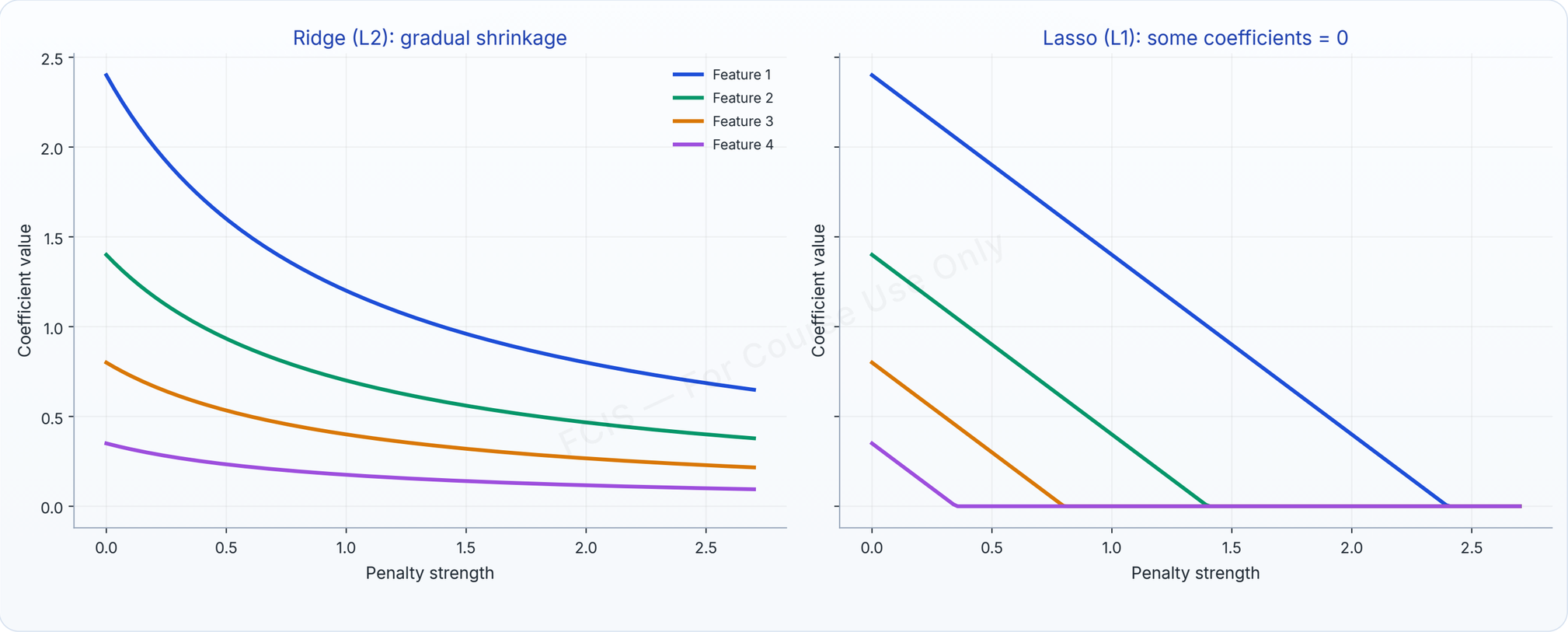
> 💡 A stronger penalty is not always better. It must balance bias and variance.

Regularization Paths

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Ridge shrinks continuously; Lasso can remove coefficients

The curves use **an illustrative orthogonal, standardized-feature case** to isolate how increasing the penalty changes coefficients.



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💡 **Takeaway.** Lasso can give sparse models; **Ridge** usually leaves nonzero coefficients. Exact paths depend on the data and on the scaling convention for the penalty.

Part VII

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Regression Evaluation

A **regression model** should be evaluated using metrics that match the prediction goal and application context.

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MAE

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Mean Absolute Error

Mean Absolute Error is:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Interpretation

MAE is the average absolute prediction error in target units.

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Advantages

- easy to understand;
- less sensitive to outliers than MSE;
- directly interpretable.

Limitation

- not smooth at zero error, though this is usually not a practical problem for evaluation.

MSE and RMSE

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Penalizing larger errors

Mean Squared Error:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Root Mean Squared Error:

$$RMSE = \sqrt{MSE}$$

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Interpretation

- MSE penalizes large errors strongly.
- RMSE returns to the original target unit.

If large errors are especially costly, **RMSE** may be more informative than MAE.

Coefficient of Determination

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R^2

The **coefficient** of determination is:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

Interpretation:

- $R^2 = 1$ means perfect prediction on the evaluated data.
- $R^2 = 0$ means no improvement over predicting the mean.
- $R^2 < 0$ means worse than predicting the mean.

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💡 A high training R^2 does not guarantee good test performance.

Choosing a Regression Metric

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The metric should match the application

Situation	Possible metric emphasis
Errors have linear cost	MAE
Large errors are especially bad	RMSE
Need variance-explained summary	R^2
Need easy unit interpretation	MAE or RMSE
Comparing across different targets	normalized or relative metrics

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Example

In medical dosage prediction, a few very large errors may be dangerous. **RMSE** or application-specific safety metrics may be important.

In rent estimation, MAE may be easy to explain:

> “The model is off by about 350 SAR on average.”

Evaluation Workflow

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Train, validate, test

A good workflow:

1. Fit preprocessing on the training set.
2. Train the model on training data.
3. Tune hyperparameters using validation data.
4. Compare models using validation performance.
5. Evaluate once on the test set.
6. Report the metric and interpret it in context.

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💡 Evaluation is part of the modeling process, not an afterthought.

Part VIII

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Worked Mini-Case

Let us connect the full regression pipeline using a small apartment-rent example.

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Mini-Case: Apartment Rent

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Problem formulation

A real-estate website wants to estimate **apartment rent**.

Features

Feature	Type	Preprocessing
Area	numerical	scaling
Number of rooms	discrete numerical	scaling or keep as count
District	nominal categorical	one-hot encoding
Building age	numerical	scaling, possible nonlinear term
Furnished	binary	encode 0/1

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Target

$$y = \text{monthly rent}$$

> This is a supervised **regression problem**.

Mini-Case: Candidate Models

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Different choices, different assumptions

Model	Assumption / behavior
Linear regression	rent changes linearly with features
Polynomial features + linear regression	allows curvature
Ridge regression	linear model with coefficient shrinkage
Lasso regression	linear model with possible feature selection

Evaluation plan

- Use validation data to select feature transformations and λ.
- Use test data only once for final evaluation.
- Report MAE and RMSE.
- Interpret whether the error is acceptable for users.

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Mini-Case: Worked Validation Comparison

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Decide using evidence, not training fit

Suppose the apartment-rent pipeline is fitted on training data, with model choices and hyperparameters selected by **validation**. These are **hypothetical classroom results**, not measurements from a real dataset.

Candidate model	Validation MAE (SAR/month)	Validation RMSE (SAR/month)	Interpretation
Linear regression	540	720	Simple baseline
Polynomial features + linear regression	465	640	Captures useful curvature
Ridge regression	**430**	580	Lowest average absolute error
Lasso regression	450	**545**	Fewer large errors by RMSE

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Example reasoning

- If typical absolute rent error matters most, **Ridge** is the first candidate to investigate.
- If larger errors are more costly, **Lasso** has the lower RMSE in this illustration.
- Do **not** use the test set to settle this decision: choose the metric in advance, select on validation, and test the chosen workflow once.

💡 A numerical comparison is evidence, not an automatic winner. Consider the metric, uncertainty in the estimates, interpretability, and deployment requirements.

Mini-Case: Reasoning Questions

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What should students explain?

1. Why is `district` not directly numerical?
2. Why should area and building age possibly be scaled?
3. Why might building age have a nonlinear effect?
4. Why might Ridge help when many district indicators are used?
5. Why might MAE be easier to communicate to end users than MSE?
6. According to the **illustrative validation table**, which model has the lowest MAE, and which has the lowest RMSE?
7. Why should the final test set not be used to choose λ or the polynomial degree?

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A good model report explains both the numerical results and the reasoning behind preprocessing and modeling choices.

Practice & Consolidation

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Regression Modeling and Common Mistakes

Apply the full learning process: calculate predictions, compare models, diagnose **overfitting**, and justify **leakage-safe evaluation**.

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Activity: Compute Predictions

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Manual practice

Suppose:

$$\widehat{\text{score}} = 30 + 3(\text{hours}) + 0.2(\text{attendance}) + 5(\text{GPA})$$

For a student with:

$$\text{hours} = 8 \qquad \text{attendance} = 90 \qquad \text{GPA} = 3.0$$

compute the predicted score.

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Questions

1. What is $\hat{y}$?
2. If the actual score is 82, what is the residual?
3. Did the model overpredict or underpredict?

Manual practice

Suppose:

$$\widehat{\text{score}} = 30 + 3(\text{hours}) + 0.2(\text{attendance}) + 5(\text{GPA})$$

For a student with:

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Questions

1. What is $\hat{y}$?
2. If the actual score is 82, what is the residual?
3. Did the model overpredict or underpredict?

$\hat{y} = 30 + 3(8) + 0.2(90) + 5(3) = 87$. Residual $= 82 - 87 = -5$, so the model overpredicted by 5.

Activity: Choose the Better Model

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Compare validation metrics

Two rent-prediction models were evaluated on validation data.

Model	MAE	RMSE	Interpretation
A	410	690	
B	460	540	

Questions

1. Which model has smaller average **absolute error**?
2. Which model has fewer very large errors?
3. Which one would you choose if large errors are very costly?
4. Why is there not always a single obvious answer?

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Compare validation metrics

Two rent-prediction models were evaluated on validation data.

Model	MAE	RMSE	Interpretation
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Questions

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> Model A has lower MAE, while Model B has lower **RMSE**. If large errors are especially costly, Model B may be preferable.

Activity: Detect Overfitting

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Training vs. validation performance

Model	Training RMSE	Validation RMSE
Linear	18.5	20.1
Degree 3 polynomial	10.2	14.0
Degree 12 polynomial	1.1	38.7

Questions

1. Which model underfits?
2. Which model likely overfits?
3. Which model appears best by validation RMSE?
4. Why should we not choose the model with the lowest training error?

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Training vs. validation performance

Model	Training RMSE	Validation RMSE
Linear	18.5	20.1
Degree 3 polynomial	10.2	14.0
Degree 12 polynomial	1.1	38.7

Questions

1. Which model underfits?
2. Which model likely overfits?
3. Which model appears best by validation RMSE?
4. Why should we not choose the model with the lowest training error?

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Degree 12 likely overfits. Degree 3 appears best by validation RMSE. The lowest **training error** is not the goal.

Common Mistakes

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Regression modeling errors

Mistake	Why it is a problem
Evaluating only training error	hides overfitting
Using target-derived features	leakage
Ignoring feature scale with regularization	penalty becomes hard to interpret
Choosing polynomial degree using test error repeatedly	test set becomes validation set
Interpreting coefficients without considering units	misleading comparisons
Removing outliers automatically	may remove important rare cases
Reporting MSE without units explanation	difficult for users to understand
Assuming linearity without checking residuals	may underfit nonlinear patterns

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Mini-Quiz

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Check your understanding

1. What is the difference between a prediction error and a loss?
2. Why does MSE penalize large errors more than MAE?
3. What does the learning rate control in gradient descent?
4. Why can polynomial features cause overfitting?
5. What is the main difference between Ridge and Lasso?
6. Why should λ be chosen using validation data?
7. Which metric is more interpretable in the original target unit: MSE or RMSE?
8. What does $R^2 < 0$ mean?

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Check your understanding

1. What is the difference between a prediction error and a loss?
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Suggested answers: a residual is the signed prediction error, a per-example loss penalizes that error, and an objective (such as MSE) aggregates losses; MSE squares errors; **learning rate** controls update size; high-degree polynomials can fit noise; Ridge shrinks, Lasso can zero coefficients; hyperparameters need validation; RMSE has original units; the model is worse than predicting the mean.

Lecture 3 Summary

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Key ideas to retain

Linear regression and optimization

- Regression predicts a **continuous target**.
- Linear regression uses:

$$\hat{y} = w^\top x + b.$$

- Residuals measure prediction errors.
- MSE is a common regression objective.
- Gradient descent updates parameters to reduce loss.
- Batch, stochastic, and mini-batch GD differ in how much data each update uses.

Generalization and evaluation

- More flexibility can reduce training error but increase overfitting.
- Polynomial features can represent nonlinear patterns.
- Ridge and Lasso add penalties to control complexity.
- MAE, MSE, RMSE, and R^2 describe regression performance differently.
- Model choice should be based on validation performance and application context.

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End-of-Lecture Check

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Five questions to test your understanding

1. How does a **linear regression** model make a prediction?
2. What does gradient descent change during training?
3. Why does scaling matter when regularization is used?
4. How can polynomial features produce both better fits and overfitting?
5. Which regression metric would you report to a non-technical stakeholder, and why?

Preparation for next lecture: Review the sigmoid function, probability interpretation, binary classification, and the difference between regression and classification.

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Suggested Reading

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Primary and supporting references

Primary reference

G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning with Applications in Python*, Springer, 2023.

- Review chapters/sections on linear regression, model flexibility, training/test error, and regularization.

Supporting references

A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly, 2022.

- Review linear models, gradient descent, polynomial regression, and regularized linear models.

K. P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022.

- Use selected sections for a deeper probabilistic view of regression, loss functions, and optimization.

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💡 When reading, focus on how prediction, loss, optimization, and generalization form one connected learning process.